

Отчет по лабораторной работе № 4

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1 Цель работы

Получение навыков правильной работы с репозиториями git.

2 Задание

1. Выполнить работу для тестового репозитория.
2. Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

3 Теоретическое введение

Gitflow Workflow опубликована и популяризована Винсентом Дриссенем. Gitflow Workflow предполагает выстраивание строгой модели ветвления с учётом выпуска проекта. Данная модель отлично подходит для организации рабочего процесса на основе релизов. Работа по модели Gitflow включает создание отдельной ветки для исправлений ошибок в рабочей среде.

Семантическое версионирование описывается в манифесте семантического версионирования. Кратко его можно описать следующим образом: Версия задаётся в виде кортежа МАЖОРНАЯ_ВЕРСИЯ.МИНОРНАЯ_ВЕРСИЯ.ПАТЧ. Номер версии следует увеличивать: МАЖОРНУЮ версию, когда сделаны обратно несовместимые изменения API. МИНОРНУЮ версию, когда вы добавляете новую функциональность, не нарушая обратной совместимости. ПАТЧ-версию, когда вы делаете обратно совместимые исправления. Дополнительные обозначения для предрелизных и билд-метаданных возможны как дополнения к МАЖОРНАЯ.МИНОРНАЯ.ПАТЧ формату.

Спецификация Conventional Commits: Соглашение о том, как нужно писать сообщения commit'ов. Совместимо с SemVer. Даже вернее сказать, сильно связано с семантическим версионированием. Регламентирует структуру и основные типы коммитов.

4 Выполнение лабораторной работы

4.1 Установка ПО. Установка git-flow, Node.js.

Настройка Node.js. Общепринятые коммиты

1. Устанавливаем git-flow (рис. fig. 4.1).

```
macos@macos:~$ cd /tmp
macos@macos:~/tmp$ wget -O gitflow-certifcate.py https://raw.githubusercontent.com/petervandendries/gitflow/develop/contrib/gitflow-installer.sh
macos@macos:~/tmp$ chmod +x gitflow-installer.sh
macos@macos:~/tmp$ ./gitflow-installer.sh install stable
## git-flow no more installer. Ask
Installing git-flow to /usr/local/bin
Cloning from /usr/local/bin to gitflow
Cloning into 'gitflow'...
remote: Enumerating objects: 4278, done.
remote: Total 4278 (delta 815, reused 0 (delta 0), push-reused 4278 (from 1))
Receiving objects: 100% (4278/4278), 3.74 MiB | 1.48 MiB/s, done.
Resolving objects: 100% (4278/4278), done.
```

Рис. 4.1: Установка git-flow

2. Устанавливаем Node.js (рис. fig. 4.2). (рис. fig. ??).

```
macos@macos:~$ sudo apt install nodejs
Last metadata expiration check: 0:04:20 ago on Mon 24 Feb 2020 08:22:22 AM EST.
Dependencies resolved.
Package Architecture Version Repository Size
Installing nodejs amd64 12.0.0-1~focal updates 52 k
Installing dependencies: nodejs amd64 12.0.0-1~focal updates 10 k
Installing main dependencies: nodejs amd64 12.0.0-1~focal updates 10 k
```

Рис. 4.2: Установка Node.js

```
macos@macos:~$ sudo apt install python
Last metadata expiration check: 0:05:07 ago on Mon 24 Feb 2020 08:22:22 AM EST.
Dependencies resolved.
Package Architecture Version Repository Size
Installing python amd64 3.8.0-1~focal updates 2.8 k
Transaction Summary
Install: 1 package
Total space: 2.8 k
```

3. Для работы с Node.js добавим каталог с исполняемыми файлами, устанавливаемыми уагн, в переменную PATH. (рис. fig. 4.3).

```
home@ubuntu:~$ npm init
added new lines to /home/tobias/.bashrc
next configuration changes were made:
export PATH="/home/tobias/.local/share/npm"
case "$PATH" in
*) export PATH="$PATH:$HOME/.npm" ;;
*)
done
To start using npm, run:
source /home/tobias/.bashrc
home@ubuntu:~$ source ~/.bashrc
bash: /home/tobias/.bashrc: No such file or directory
home@ubuntu:~$ source ~/.bashrc
tobias@ubuntu:~$ npm add -g commitizen
update available! 0.0.0 -> 0.4.1
Changelog: https://github.com/commitizen/commitizen/releases/tag/v0.4.1
Run "npm add -g commitizen" to update.
Follow @commitizen for updates: https://twitter.com/commitizen

home@ubuntu:~$ npm install commitizen --save-dev
added to package.json: commitizen@0.0.0
packages: +11
Progress: resolved 11, reused 0, downloaded 11, added 11, done
/home/tobias/.local/share/npm/global/0
commitizen 0.0.0
Done in 3.7s
```

Рис. 4.3: Настройка Node.js

4. Настраиваем commitizen, standard-changelog. (рис. fig. 4.4). (рис. fig. 4.5).

```
home@ubuntu:~$ npm add -g commitizen
update available! 0.0.0 -> 0.4.1
Changelog: https://github.com/commitizen/commitizen/releases/tag/v0.4.1
Run "npm add -g commitizen" to update.
Follow @commitizen for updates: https://twitter.com/commitizen

home@ubuntu:~$ npm install commitizen --save-dev
added to package.json: commitizen@0.0.0
packages: +11
Progress: resolved 11, reused 0, downloaded 11, added 11, done
/home/tobias/.local/share/npm/global/0
commitizen 0.0.0
Done in 3.7s
```

Рис. 4.4: commitizen

```
home@ubuntu:~$ npm add -g standard-changelog
added to package.json: standard-changelog@0.0.0
packages: +11
Progress: resolved 11, reused 0, downloaded 11, added 11, done
/home/tobias/.local/share/npm/global/0
standard-changelog 0.0.0
Done in 3.7s
```

Рис. 4.5: standard-changelog

4.2 Практический сценарий использования git.

Создание репозитория. Работа с репозиторием.

5. Создаем репозиторий на GitHub. Для примера назовём его git-extended. Делаем первый коммит и выкладываем на github: (рис. fig. 4.6).

```
tvbondar@tvbondar:~/git-extended$ git add
tvbondar@tvbondar:~/git-extended$ git commit -m "first commit"
[master (root)] first commit
1 file changed, 1 deletion(-)
tvbondar@tvbondar:~/git-extended$ git remote add origin git@github.com:tvbondar/git-extended.git
error: remote origin already exists.
tvbondar@tvbondar:~/git-extended$ git remote -v
origin  git@github.com:tvbondar/git-extended.git (fetch)
origin  git@github.com:tvbondar/git-extended.git (push)
tvbondar@tvbondar:~/git-extended$ git remote add git@github.com:tvbondar/git-extended.git
usage: git remote add <specname> <url>

--if, --no[!fetch]  fetch the remote branches
--no[!]tags        import all tags and associated objects when fetching
                   or do not fetch any tag at all (--no-tags)
--[no-]track-branches  branches to track
--[no-]master-branch  master branch
--[no-]mirror[!push|fetch]  set up remote as a mirror to push to or fetch from

tvbondar@tvbondar:~/git-extended$ git push -u origin master
error: src refspec master does not match any
tvbondar@tvbondar:~/git-extended$ git push -u tvbondar:git-extended.git
tvbondar@tvbondar:~/git-extended$ git push -u master
fatal: 'master' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.

tvbondar@tvbondar:~/git-extended$ git push -u
Enumerating objects: 5, done.
Writing objects: 100% (5/5), 240.00 KiB, done.
remote: (error) repository not found: tvbondar:git-extended.git
remote: (error) repository not found: tvbondar:git-extended.git
branch 'master' set up to track 'origin/master'.
```

Рис. 4.6: git-extended

6. Конфигурация общепринятых коммитов. Для этого добавим в файл package.json команду для формирования коммитов: (рис. fig. 4.7).

```
/home/tvbondar/git-extended/package.json
{
  "name": "git-extended",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:tvbondar/git-extended.git",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "Tatiana Bondar sinopahfox@yandex.ru",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рис. 4.7: Конфигурация коммитов

7. Добавляем файлы, выполняем коммит, отправляем на Github. (рис. fig. 4.8).

```

tvbondar@fedora:~/git-extended$ git add .
tvbondar@fedora:~/git-extended$ git cz
Parsing JSON at /home/tvbondar/git-extended/package.json for commitizen config failed:

tvbondar@fedora:~/git-extended$ mc

tvbondar@fedora:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

Select the type of change that you're committing:
> revert: Reverts a previous commit
feat: A new feature
fix: A bug fix
docs: Documentation only changes
style: Changes that do not affect the meaning of the code (white-space, formatting, missing semi-colons, etc)
refactor: A code change that neither fixes a bug nor adds a feature
perf: A code change that improves performance
(Move up and down to reveal more choices)
tvbondar@fedora:~/git-extended$ git add .
tvbondar@fedora:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

Select the type of change that you're committing: feat: A new feature
What is the scope of this change (e.g. component or file name): (press enter to skip)
Write a short, imperative tense description of the change (max 94 chars):
(19) change_package.json
Provide a longer description of the change: (press enter to skip)

Are there any breaking changes? No
Does this change affect any open issues? No
[main c7366dd] feat: change_package.json
1 file changed, 18 insertions(+)
create mode 100644 package.json
tvbondar@fedora:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 573 bytes | 573.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:tvbondar/git-extended.git
563b201..c7366dd main -> main
tvbondar@fedora:~/git-extended$

```

Рис. 4.8: Отправка изменений на github

8. Инициализируем git-flow Префикс для ярлыков установим в v. Проверьте, что Вы на ветке develop: Загрузите весь репозиторий в хранилище: Установите внешнюю ветку как вышестоящую для этой ветки: Создадим релиз с версией 1.0.0 (рис. fig. 4.9). (рис. fig. 4.10). (рис. fig. 4.11).

```

tvbondar@fedora:~/git-extended$ git flow init
Which branch should be used for bringing forth production releases?
- main
Branch name for production releases: [main]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/tvbondar/git-extended/.git/hooks]
tvbondar@fedora:~/git-extended$

```

Рис. 4.9: Часть 1

```

tvbondar@fedora:~/git-extended$ git branch
* develop
main
tvbondar@fedora:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote: https://github.com/tvbondar/git-extended/pull/new/develop
remote:
To github.com:tvbondar/git-extended.git
* [new branch] develop -> develop
tvbondar@fedora:~/git-extended$

```

Рис. 4.10: Часть 2

```

tvbondar@fedora:~/git-extended$ git branch --set-upstream-to=origin/develop develop
branch 'develop' set up to track 'origin/develop'.
tvbondar@fedora:~/git-extended$ git flow release start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'

```

Рис. 4.11: Часть 3

9. Создадим журнал изменений Добавим журнал изменений в индекс Зальём релизную ветку в основную ветку Отправим данные на github Создадим релиз на github. Для этого будем использовать утилиты работы с github: (рис. fig. 4.12). (рис. fig. 4.13). (рис. fig. 4.14). (рис. fig. 4.15).

```

tvbondar@fedora:~/git-extended$ yarn global add standard-changelog
yarn global v1.22.22
[1/4] Resolving packages...
[2/4] Fetching packages...
[3/4] Linking dependencies...
[4/4] Building fresh packages...
success Installed "standard-changelog@6.0.0" with binaries:
  - standard-changelog
Done in 8.44s.
tvbondar@fedora:~/git-extended$ npx standard-changelog
Need to install the following packages:
standard-changelog@6.0.0
Ok to proceed? (y) y

✓ created CHANGELOG.md
✓ output changes to CHANGELOG.md
tvbondar@fedora:~/git-extended$

```

Рис. 4.12: Часть 1

```

tvbondar@fedora:~/git-extended$ npx standard-changelog
Need to install the following packages:
standard-changelog@6.0.0
Ok to proceed? (y) y

✓ created CHANGELOG.md
✓ output changes to CHANGELOG.md
tvbondar@fedora:~/git-extended$ ls
CHANGELOG.md package.json README.md
tvbondar@fedora:~/git-extended$ standart-changelog --first-release
bash: standart-changelog: command not found...
tvbondar@fedora:~/git-extended$ git add CHANGELOG.md
tvbondar@fedora:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 121e622] chore(site): add changelog
1 file changed, 10 insertions(+)
create mode 100644 CHANGELOG.md
tvbondar@fedora:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 10 ++++++++
1 file changed, 10 insertions(+)
create mode 100644 CHANGELOG.md
Already on 'main'
Your branch is ahead of 'origin/main' by 2 commits.
(use "git push" to publish your local commits)
fatal: no tag message?
Fatal: Tagging failed. Please run finish again to retry.
tvbondar@fedora:~/git-extended$ git flow release finish 1.0.0
Branches 'main' and 'origin/main' have diverged.
And local branch 'main' is ahead of 'origin/main'.
Already on 'main'
Your branch is ahead of 'origin/main' by 2 commits.
(use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 10 ++++++++
1 file changed, 10 insertions(+)
create mode 100644 CHANGELOG.md
Deleted branch release/1.0.0 (was 121e622).

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'main'
- The release was tagged 'v1.0.0'
- Release tag 'v1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

tvbondar@fedora:~/git-extended$

```

Рис. 4.13: Часть 2

```

tvbondar@fedora:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 691 bytes | 691.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To github.com:tvbondar/git-extended.git
   c7366dd..159ff27  develop -> develop
   c7366dd..f33b639  main -> main
tvbondar@fedora:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 169 bytes | 169.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:tvbondar/git-extended.git
 * [new tag]         v1.0.0 -> v1.0.0
tvbondar@fedora:~/git-extended$

```

Рис. 4.14: Часть 3

```

tvbondar@fedora:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/tvbondar/git-extended/releases/tag/v1.0.0
tvbondar@fedora:~/git-extended$

```

Рис. 4.15: Часть 4

10. Создадим ветку для новой функциональности. Следующим шагом следует объединить ветку `feature_branch` с `develop`. (рис. fig. 4.16).

```

tvbondar@fedora:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

tvbondar@fedora:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was 159ff27).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

tvbondar@fedora:~/git-extended$

```

Рис. 4.16: Создание и слияние веток

11. Создадим релиз с версией 1.2.3. Обновим номер версии в файле `package.json`. Установим её в 1.2.3. (рис. fig. 4.17). (рис. fig. 4.18).

```

tvbondar@fedora:~/git-extended$ git flow release start 1.2.3
Switched to a new branch 'release/1.2.3'

Summary of actions:
- A new branch 'release/1.2.3' was created, based on 'develop'
- You are now on branch 'release/1.2.3'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.2.3'

tvbondar@fedora:~/git-extended$

```

Рис. 4.17: Релиз v1.2.3.

```

package.json  [-M--] 19 L: [ 1+ 2 3/ 19] *(47 / 441b) 0034 0x022
{
  "name": "git-extended",
  "version": "1.2.3",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:tvbondar/git-extended.git",
  "scripts": {
    "test": "echo \\\"Error: no test specified\\\" && exit 1"
  },
  "keywords": [],
  "author": "Tatiana Bondar sinopahfox@yandex.ru",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}

```

Рис. 4.18: Изменение package.json

12. Создадим журнал изменений Добавим журнал изменений в индекс Зальём релизную ветку в основную ветку Отправим данные на github Создадим релиз на github с комментарием из журнала изменений: (рис. fig. 4.19). (рис. fig. 4.20). (рис. fig. 4.21).

```

tvbondar@fedora:~/git-extended$ npx standard-changelog
✓ output changes to CHANGELOG.md
tvbondar@fedora:~/git-extended$

```

Рис. 4.19: Часть 1

```

tvbondar@fedora:~/git-extended$ git add CHANGELOG.md
tvbondar@fedora:~/git-extended$ git commit -am 'chore(site): update changelog'
[release/1.2.3 5f6603d] chore(site): update changelog
 2 files changed, 5 insertions(+), 1 deletion(-)
tvbondar@fedora:~/git-extended$ git flow release finish 1.2.3
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Merge made by the 'ort' strategy.
  CHANGELOG.md | 4 ++++
  package.json | 2 +-
  2 files changed, 5 insertions(+), 1 deletion(-)
Already on 'main'
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
  CHANGELOG.md | 4 ++++
  package.json | 2 +-
  2 files changed, 5 insertions(+), 1 deletion(-)
Deleted branch release/1.2.3 (was 5f6603d).

Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'main'
- The release was tagged 'v1.2.3'
- Release tag 'v1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

tvbondar@fedora:~/git-extended$

```

Рис. 4.20: Часть 2

```

tvbondar@fedora:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 4 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 759 bytes | 759.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
To github.com:tvbondar/git-extended.git
   159ff27..f882a67  develop -> develop
   f33b639..e980d10  main -> main
tvbondar@fedora:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 172 bytes | 172.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:tvbondar/git-extended.git
 * [new tag]         v1.2.3 -> v1.2.3
tvbondar@fedora:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/tvbondar/git-extended/releases/tag/v1.2.3
tvbondar@fedora:~/git-extended$

```

Рис. 4.21: Часть 3

5 Выводы

Мы получили навыки правильной работы с репозиториями git.

Список литературы

Рабочий процесс с Gitflow(электронный ресурс) URL: <https://yamadharma.github.io/ru/post/2021-01-01-gitflow-workflow/>